The PCIR Protocol: A Unified Information-Theoretic Framework for Physics

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Abstract

Current theoretical physics is fragmented into distinct domains: General Relativity (geometry), Quantum Mechanics (probability), and Thermodynamics (time). This executive summary presents the **PCIR (Parsimony, Context, Information, Reflexion)** framework, a unified theory positing that physical laws are emergent properties of a self-evidencing system. By demonstrating that the physical metric tensor is isomorphic to the Fisher Information Metric, we establish that Spacetime, Quantum Dynamics, and the Thermodynamic Arrow of Time are the geometric, statistical, and metabolic manifestations of efficient information processing.

1 The Geometric Basis: Parsimony (P) and Context (C)

Source Paper: The Geometry of Parsimony: Spacetime as a Statistical Manifold equipped with a Fisher Information Metric

We formalize the "Emergent Gravity" program using Information Geometry, treating spacetime not as fundamental, but as arising from the entanglement structure of underlying quantum degrees of freedom.

- **The Metric Isomorphism:** We propose that the physical metric tensor $g_{\mu\nu}$ is isomorphic to the Fisher Information Metric (FIM) defined on a statistical manifold of probability distributions. This metric measures the distinguishability of neighboring states.
- **Parsimony as Curvature:** The principle of "Parsimony"—the drive to minimize model complexity—is geometrically realized as the flattening of the statistical manifold. We demonstrate that the minimization of Kullback-Leibler divergence leads directly to the minimization of the Ricci scalar curvature. The Einstein-Hilbert action is reinterpreted as a measure of information density.
- **Context as Boundary:** We identify the system's "Context" with a holographic boundary or Markov Blanket. By associating the variation of information on this boundary with the stress-energy tensor $T_{\mu\nu}$, we recover the Einstein Field Equations from purely information-theoretic constraints.

2 The Microscopic Dynamics: Information (I)

Source Paper: Epistemic Dynamics: Deriving the Schrödinger Equation from the Principle of Least Surprise

We resolve interpretational ambiguities in quantum mechanics by adopting a "QBism" (Quantum Bayesianism) stance, where the wavefunction represents an agent's subjective degrees of belief rather than physical reality.

- **Kinetic Energy as Fisher Information:** We demonstrate that the kinetic energy term in the Schrödinger equation is isomorphic to the Fisher Information of the probability density. High Fisher Information implies high localization, which represents high complexity or "Parsimony cost".
- **Dynamics as Inference:** The time-evolution of the wavefunction is modeled as a diffusion process on a statistical manifold. We derive the Schrödinger equation from a variational principle where the system minimizes a functional composed of Fisher Information (Parsimony) and Expected Energy (Accuracy).
- **Collapse as Updating:** Wavefunction collapse is identified as Bayesian updating. It is the discontinuity of a probability distribution when new information is acquired, representing an update of knowledge rather than a physical contraction.

3 The Thermodynamic Driver: Reflexion (R)

Source Paper: Renormalization Group Flow as Dissipative Gradient Descent: A Thermodynamic Basis for Model Selection

We address the emergence of time and the thermodynamic cost of information processing, completing the dynamic description of the PCIR framework.

- **Reflexion Defined:** We term "Reflexion" as the process where a system recursively updates its own parameters to minimize variational free energy.
- **RG Flow as Gradient Descent:** We propose that Renormalization Group (RG) flow is a gradient descent process on a Variational Free Energy landscape defined over the manifold of model parameters. The beta functions $\beta(g)$ act as gradient vectors minimizing this energy.
- **The Thermodynamic Arrow of Time:** By invoking Landauer's Principle, we argue that the "coarse-graining" of degrees of freedom (information erasure) generates a lower bound on heat dissipation. The flow toward Infrared (IR) fixed points is isomorphic to a thermodynamic relaxation process, providing a thermodynamic arrow of time driven by the metabolic cost of model simplification.

4 Conclusion

The PCIR framework provides a rigorous definition for the unification of physics:

- **Parsimony (P):** The Einstein-Hilbert Action (Gravity).
- **Context (C):** The Boundary Terms (Holography).
- **Information (I):** The Schrödinger Equation (Quantum Dynamics).
- **Reflexion (R):** Renormalization Group Flow (Thermodynamics/Time).

Together, these papers suggest that Geometry, Time, and Quantum States are emergent features of a single information-processing principle.